# Exercise 13. Recognizing Single-peakedness

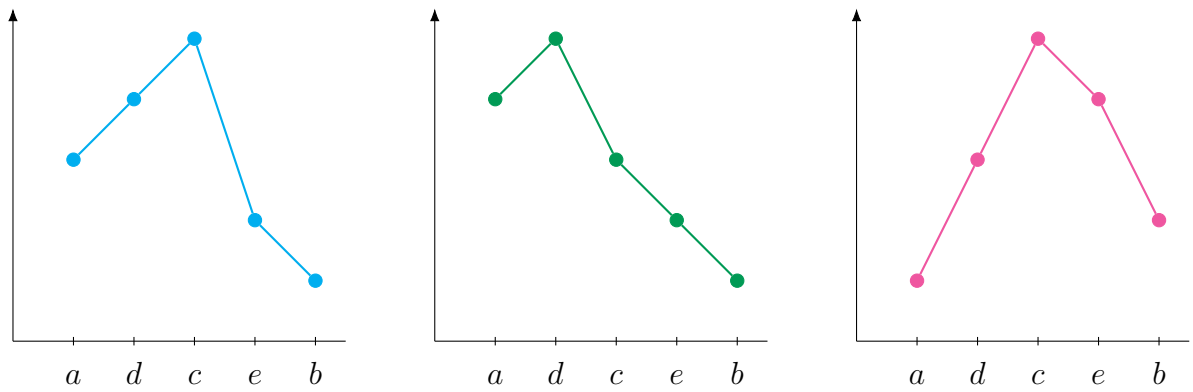
## (a)

### (*i*)

The algorithm yields the following result.

| $B$ | $L$ | $R$ | $\ell$ | $r$ | current order |
|---|---|---|---|---|---|
| $\{a, b\}$ | $\emptyset$ | $\emptyset$ | $z_l$ | $z_r$ | $z_l > a > \cdots > b > z_r$ |
| $\{d, e\}$ | $\emptyset$ | $\{e\}$ | $a$ | $b$ | $z_l > a > d > \cdots > e > b > z_r$ |
| $\{c\}$ | $\emptyset$ | $\{c\}$ | $d$ | $e$ | $z_l > a > d > c > e > b > z_r$ |

When we plot the preferences according to the discovered order we get the following picture.



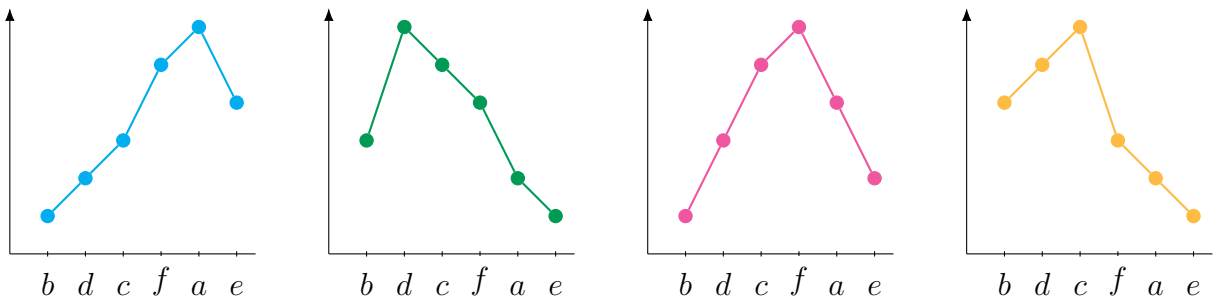

Voter 1 Voter 2 Voter 3

One can observe, that the preferences are single peaked.

***(ii)***

| $B$ | $L$ | $R$ | $\ell$ | $r$ | current order |
|---|---|---|---|---|---|
| $\{b, e\}$ | $\emptyset$ | $\emptyset$ | $z_l$ | $z_r$ | $z_l > b > \cdots > e > z_r$ |
| $\{d, a\}$ | $\{d\}$ | $\{a\}$ | $b$ | $e$ | $z_l > b > d > \cdots > a > e > z_r$ |
| $\{c, f\}$ | $\{c\}$ | $\{f\}$ | $d$ | $a$ | $z_l > b > d > c > f > a > e > z_r$ |

When we plot the preferences according to the discovered order we get the following picture.




Voter 1 Voter 2 Voter 3 Voter 4 & 5

One can observe, that the preferences are single peaked.

***(iii)***

| $B$ | $L$ | $R$ | $\ell$ | $r$ | current order |
|---|---|---|---|---|---|
| $\{e, f\}$ | $\emptyset$ | $\emptyset$ | $z_l$ | $z_r$ | $z_l > e > \cdots > f > z_r$ |
| $\{c, d\}$ | $\{d\}$ | $\{c\}$ | $e$ | $f$ | $z_l > e > d > \cdots > c > f > z_r$ |
| $\{a, b\}$ | $\{\}$ | $\{a, b\}$ | $d$ | $c$ | $z_l > e > d > > \begin{matrix} b \\ a \end{matrix} > c > f > z_r$ |

Since in the last iteration of the algorithm the set $R$ has size 2, there is no linear order, for which the given preference profile is single-peaked.

## (b)

W.o.l.g we can assume a preference profile of the following form

| |
|---|
| 1 |
| $x_1$ |
| $x_2$ |
| $\vdots$ |
| $x_m$ |

This preference profile will be called $[R_N]_m$.

We prove by induction, that $[R_N]_m$ is single-peaked with respect to at least $2^{m-1}$ orders for all $m \in \mathbb{N} \setminus \{0\}$.

**Base case:**

For $m = 1$, $[R_N]_m = [R_N]_1$ is single-peaked w.r.t the order $x_1$.

**Inductive Step:**

Let $[R_N]_{m-1}$ be single-peaked with respect to at least $2^{m-1}$ orders $\forall m \in \mathbb{N} \setminus \{0\}$ contained in the set $O_{[R_N]_{m-1}}$.

In general the set $O_{[R_N]_m}$ contains all orders with which $[R_N]_m$ is single-peaked.

Now we choose an arbitrary order from $O_{[R_N]_{m-1}}$.

Now the profile $[R_N]_{m+1}$ will be single-peaked for the orders

$$x_{m+1} > x_1 > \ldots > x_m$$

$$x_1 > \ldots > x_m > x_{m+1}$$

This works since $x_{m+1}$ is the lowest of the ranked candidates in $[R_N]_{m+1}$.

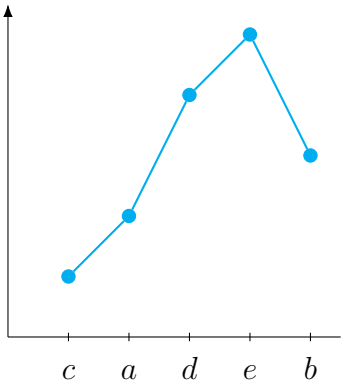
Since this order was picked arbitrarily the reasoning holds for all elements of $O$. Thus, the number of orderings doubled relative to $[R_N]_m$.

If we assume $|O_{[R_N]_{m-1}}| = 2^{m-1}$, then we can conclude $|O_{[R_N]_m}| = 2^m$.

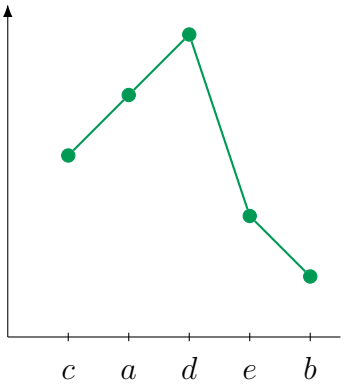
# Exercise 14. Single-peaked, single-caved and value restricted preferences

**(*i*)**

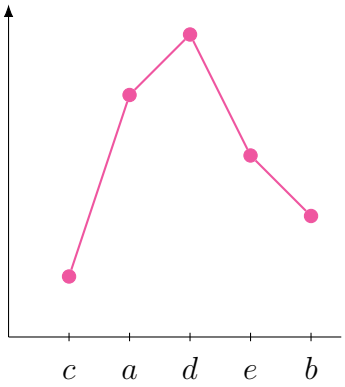
The order, for which the first preference profile is single-peaked is $c > a > d > e > b$.




Voter 1

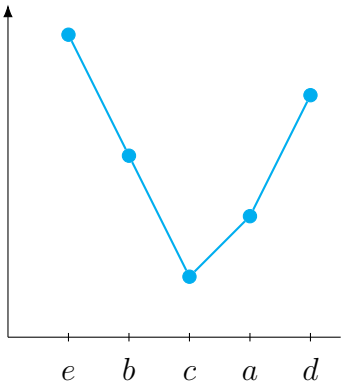



Voter 2

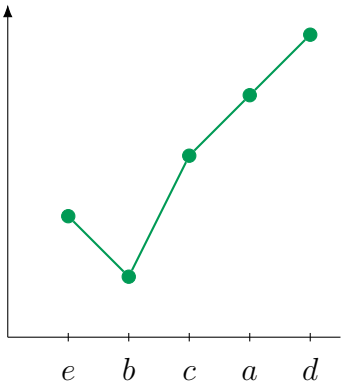



Voter 3

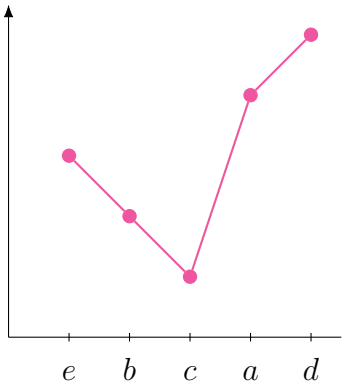
This preference profile is single-caved with the following order: $e > b > c > a > d$




Voter 1




Voter 2




Voter 3

Since the first preference profile is single-peaked, we know it is also value restricted since $D_{\text{SP}} \subseteq D_{\text{VR}}$ as slide 87 of the lecture states.

***(ii)***

This preference profile is not single-peaked.

In the first step of the algorithm the order $a > \ldots > d$ emerges. In the second step $c$ has to be placed next to $d$ and $a$.

So the intersection of $R$ and $L$ is not empty and there is no order for which the second preference profile will be single-peaked.

The second preference profile is not single-caved, since the set of candidates ranked as first is $\{a, b, c, d\}$, which size is not $\leq 2$.

It is, however value-restricted. Consider the following preference profiles

| 1 | 2 | 2 |
|---|---|---|
| $a$ | $c$ | $b$ |
| $b$ | $b$ | $c$ |
| $c$ | $a$ | $a$ |

$R_N\big|_{\{a,b,c\}}$

| 1 | 2 | 1 | 1 |
|---|---|---|---|
| $a$ | $d$ | $b$ | $b$ |
| $b$ | $b$ | $a$ | $d$ |
| $d$ | $a$ | $d$ | $a$ |

$R_N\big|_{\{a,b,d\}}$

| 1 | 2 | 1 | 1 |
|---|---|---|---|
| $a$ | $d$ | $c$ | $c$ |
| $c$ | $c$ | $a$ | $d$ |
| $d$ | $a$ | $d$ | $a$ |

$R_N\big|_{\{a,c,d\}}$

| 2 | 1 | 1 | 1 |
|---|---|---|---|
| $b$ | $d$ | $d$ | $c$ |
| $c$ | $c$ | $b$ | $b$ |
| $d$ | $b$ | $c$ | $d$ |

$R_N\big|_{\{b,c,d\}}$

For each of those preference profiles we can give an element, which satisfies a condition such that $R_N$ will be value-restricted.

- for $R_N\big|_{\{a,b,c\}}$ it holds that $a$ is never in the middle.
- for $R_N\big|_{\{a,b,d\}}$ it holds that $b$ is never the worst.
- for $R_N\big|_{\{a,c,d\}}$ it holds that $c$ is never the worst.
- for $R_N\big|_{\{b,c,d\}}$ it holds that $d$ is never in the middle.

We can conclude that $R_N$ is value-restricted.

# Exercise 15. Single-caved preferences

## (a)

No it is not.

Consider the following preference profile

| |
|---|
| 1 |
| $x \in \{a, b\}$ |
| $y \in \{a, b\} \setminus \{x\}$ |
| $c$ |

Someone might give us the cave $c$, but both $a\,P\,b$ and $b\,P\,a$ are possible (of course not at the same time).

If $a\,P\,b$ holds, then $a$ is the Condorcet-winner, else $b$ is the Condorcet-winner.

So we don't have enough information to decide whether $a$ or $b$ is the Condorcet-winner, even if we know that $c$ is the cave.

## (b)

This statement is false for $m > 2$.

As shown in a previous exercise only the 'ends' of given linear order can be Condorcet-winners.

W.o.l.g. we can assume the order to be $o := x_1 > x_2 > \ldots > x_m$

Then only the candidates $x_1$ and $x_m$ can be Condorcet-winners, since the preference profile has to be single-caved.

Thus, all candidates $x_2, x_3, \ldots, x_{m-1}$ cannot be Condorcet-winners.